

FireEye: An IoT-Based Fire Alarm and Detection System for Enhanced Safety

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Abstract—Ensuring fire safety is a critical aspect of maintaining a safe home environment. A reliable and efficient fire alarm system is essential for detecting and alerting occupants of a potential fire hazard quickly. However, many existing fire detection systems are costly and complicated, which can discourage homeowners from investing in them. To address this issue, FireEye, an IoT-based fire alarm and detection system, has been developed. The system prioritizes performance, ease of use, functionality, and affordability to provide homeowners with an accessible and reliable solution for fire safety. FireEye incorporates cutting-edge technology to deliver real-time alerts and accurate detection of fires, ensuring that homeowners can enjoy peace of mind knowing their loved ones and property are protected.

Index Terms—Fire detection, IOT, cloud [1], mobile application

I. INTRODUCTION

Fire safety is a critical concern in any building, and an effective fire alarm system can play a crucial role in protecting life and property. Fires can spread quickly and can be difficult to control, leading to injuries, fatalities, and significant damage to buildings and infrastructure. Additionally, fires can cause financial and emotional distress to those affected, and they can have long-lasting effects on the environment. By being cautious and taking steps to prevent fires, individuals can reduce the risks associated with this hazard and ensure the safety of themselves and those around them. According to the National Fire Protection Association, in 2019, there were approximately 1.3 million fires reported in the US, resulting in 3,704 civilian deaths, 16,600 civilian injuries, and \$14.8 billion in property damage [2]. In this project, we have developed a fire alarm system that is equipped with advanced sensors, communication technologies, and intelligent algorithms to detect, respond, and mitigate the risk of fire.

Our system is designed to automatically detect the presence of fire and alert the building occupants through a sound buzzer and notifications to devices such as mobile phones and computers. The system can also automatically activate a water or gas-based extinguishing system to suppress the fire and minimize property damage. The development of this system is based on the integration of various technologies, including micro-controllers, sensors, wireless communication, mobile application, firebase [3]. Our system uses a combination of flame sensors to detect the presence of fire. It is also designed to be easy to install, operate, and maintain, making it an ideal solution for both residential and commercial applications. It can be customized to meet specific needs and requirements, and it can be integrated with other building automation systems to enhance its functionality. The following report provides a comprehensive overview of our fire alarm system, including its design, development, and testing. We describe in detail the hardware and software components of the system, its operation, and its performance under various scenarios.

II. LITERATURE REVIEW

Tri Listyorini has developed a novel gadget utilizing fuzzy logic and the Internet of Things (IoT) to detect fire hotspots in Riau's peatlands, Indonesia. The device, powered by the WE-MOS ESP8266 microprocessor, effectively coordinates servo motors, buzzers, fire and temperature sensors, and security cameras. Fuzzy logic determines flame intensity, minimizing false alarms. Testing has shown that this IoT-connected fire detection device expedites monitoring while reducing false alarms. It has the potential to revolutionize fire detection in critical areas like peatlands and serves as an educational tool for high school students studying computer engineering and networking. This innovation combines environmental signif-

icance with practical application, providing valuable insights into IoT technology and fuzzy logic for future engineers and technologists [4].

Arpit Jadon proposes a novel neural network called "FireNet" to overcome the performance-size compromise in existing solutions. FireNet is a lightweight network that outperforms previous alternatives, aiming to be deployed on embedded platforms like Raspberry Pi. Extensive evaluations using conventional and customized fire datasets demonstrate promising results. The project article explores FireNet's capabilities, incorporating Convolutional Neural Networks, Embedded Systems, Fire Detection, Internet of Things (IoT), Neural Networks, and Smoke Detection. This comprehensive examination highlights FireNet's potential in bridging the gap between efficient model design and high-performance outcomes in fire detection and embedded systems [5].

Abdullah H. Altowaijri presents a novel fire detection approach that upholds high detection accuracy while prioritizing privacy. Instead of transmitting complete video footage, this method transmits derived features extracted from IoT device videos. It employs Convolutional Neural Networks (CNN) and binary video descriptors for feature extraction and data classification. The proposed technique outperforms existing methods using raw video data, achieving an impressive 97.5% classification accuracy. The study also demonstrates the suitability of the suggested video descriptors for real-time processing on the Raspberry Pi 4 platform, known for its average processing speed of 100 milliseconds per frame. This combination of innovative approaches enhances fire detection accuracy and enables efficient and privacy-conscious processing in IoT environments [6].

M. Trinath Basu proposes a NodeMCU-based IoT-based fire indication and monitoring system [7]. It communicates with a variety of parts, such as temperature and smoke sensors, and it uses an Arduino-connected buzzer to sound an alarm in case of a fire. Clear system status presentation is made possible by NodeMCU integration with an LCD display, and user transmission of status messages is facilitated by an Ethernet module. When a fire occurs, the system automatically detects it and immediately alerts the user via alarms delivered to an Android mobile app or website. This ground-breaking approach makes advantage of NodeMCU's capabilities to build a user-centric fire monitoring system that bridges the gap between distant alarms and real-time detection. The system's efficacy and usability are improved since its functioning continues even when the user is geographically separated [8].

Giacomo Peruzzi presents a prototype Video Surveillance Unit (VSU) that incorporates two embedded Machine Learning (ML) algorithms for timely forest fire detection [9,10]. Machine learning models run on a low-power device with input from audio samples and photos. Together, the ML models achieve 96.15 percent, 92.3 percent, 100.00 percent, and 96.0 percent accuracy, precision, recall, and F1 score, respectively. The LoRaWAN protocol smoothly transmits incident detection, ensuring prompt notification to accountable parties. The prototype VSU is recognized as a novel solution for accurate

and timely forest fire detection because to the integration of embedded ML algorithms and cutting-edge communication protocols. This enables quick and efficient actions in reaction to probable fire occurrences [11].

Xiaogeng Ren proposes a multi-information fusion-based integrated detection technology to address electric fires in environmentally-friendly buildings. By applying fuzzy logic reasoning and membership functions within the framework of fuzzy set theory, a comprehensive system model is established to tackle the complex interplay of problems and symptoms related to electric fires. The study focuses on identifying arc faults, a common cause of electrical fires in green buildings. Through multi-information fusion, arc faults are effectively detected, with simulation results validating the model's reliability. The research highlights the efficacy of this approach in preventing electrical fires, offering a more precise solution compared to traditional methods relying on single criteria. The empirical findings contribute to enhancing fire safety in green buildings by leveraging advanced fusion techniques to accurately identify and prevent potential arc-related fire hazards [12].

N N Mahzan et al 2018 J. Phys.: Conf. Ser have proposed a Arduino-based home fire alarm system with GSM module. They used The Arduino microcontroller which is connected to the fire sensor and the alarm buzzer. They had to add a GSM module which is connected to the Arduino microcontroller. In this paper they must have GSM system in order to use it at its best and there is a possibility of false alarm [13].

Jenil Dosani et al have proposed a NodeMCU based fire detection system. They described the use of NodeMCU, a microcontroller board, as the main component of the fire detector system. It is connected to a smoke sensor and a buzzer to detect and alert about the presence of smoke or fire. The NodeMCU collects data from the smoke sensor and sends it to the ThingSpeak server through the cloud system. Having a smoke sensor and fire sensor together can cause false detection in few circumstances [14].

Khalaf O. I. et al have proposed a IOT fire detection system using Arduino. They discussed the use of IoT technology, specifically temperature sensors (flame sensors) connected to an Arduino device, to detect and measure the intensity of fire outbreaks in various locations. The sensors serve as an early alarm system, sending email notifications to mobile phones, fire stations, and hospitals to alert them of any fire outbreaks and prevent significant damage. They utilized an IoT system consisting of temperature sensors (flame sensors) connected to an Arduino device to detect fire outbreaks and measure the intensity of heat generated by the fire. The sensors are connected to a central hub called a gateway, which collects and processes the data from the sensors [15]. They aimed to have lower cost which made it less secure and efficient in terms of service [16].

involved research, hardware and software design, and testing. The system utilizes a variety of sensors and actuators to detect and respond to fires in a home environment, with the option to add a gas releasing system for added protection. The software components provide real-time monitoring and notifications to ensure the homeowner can respond quickly in the event of a fire.

V. IMPLEMENTATION

FireEye is an IoT-based fire alarm and detection system designed to enhance safety in buildings. The system is designed to detect fire, activate a buzzer and LED lights, dispense water to the affected area, and notify users on their devices. The project has been completed and undergone testing and improvements to ensure its effectiveness in detecting fires and providing early warning to users. The FireEye team identified potential risks and implemented mitigation strategies, resulting in a highly effective and efficient system for enhanced safety.

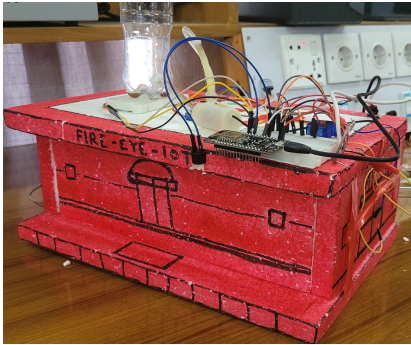


Fig. 3. Complete Project

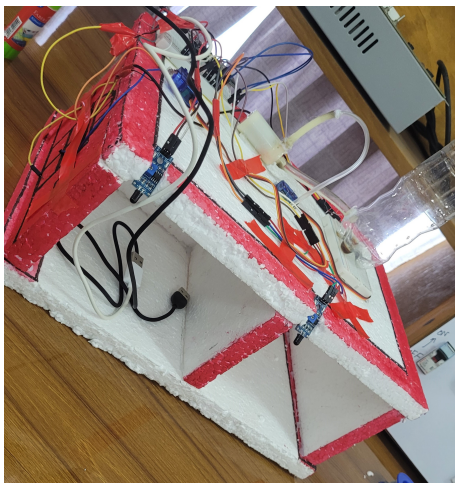


Fig. 4. Complete Project

The FireEye fire alarm and detection system is built using the NodeMCU configuration with specific pins for each of the system's components. The flame sensors are connected to D6 and D7 pins, allowing for accurate detection of fires. The

D4 and D5 pins are used to activate the buzzer and LED, respectively, providing an immediate alert to occupants of the building. The servo motor, which is used to dispense water to the affected area, is connected to D2 and D3 pins, allowing for precise control over the direction and intensity of the water flow. The water pump, which is powered individually from a separate power source, is activated when a fire is detected, ensuring the immediate suppression of the fire. This NodeMCU configuration has proven to be highly effective in detecting fires and activating the necessary responses for enhanced safety[20].

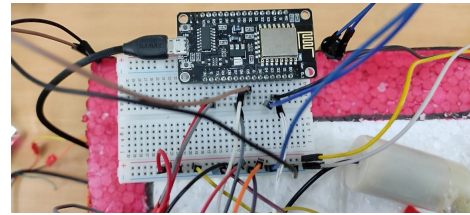


Fig. 5. NodeMCU ESP8266 configuration

In addition to the hardware components of FireEye, an Android app was developed to provide real-time notifications to users in the event of a fire. The app was built using Kotlin programming language and Android Studio development environment. Firebase was used as the database for storing and updating information related to the system's status, allowing users to receive notifications in real-time. The app provides a user-friendly interface, displaying the system's status and allowing users to configure and monitor the system's settings remotely. The app's real-time update feature ensures that users are immediately alerted in the event of a fire, enabling them to take swift and effective action to ensure their safety. Overall, the Android app is an essential component of the FireEye system, providing a crucial layer of safety and peace of mind for building occupants.

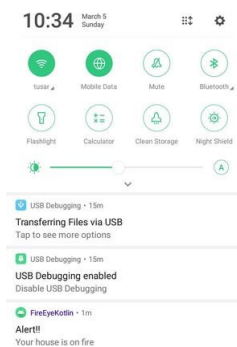


Fig. 6. Notification Sent on User Device

VI. RESULT ANALYSIS

the FireEye fire alarm and detection system is highly effective in detecting fires and providing early warning to building

occupants [21]. The system's flame sensors are highly sensitive, detecting even small fires and activating the necessary responses within seconds. The servo motor, which dispenses water to the affected area, was highly effective in controlling the direction and intensity of the water flow, suppressing the fire within seconds of activation. Our device quickly detects any fire occurrence using a highly effective sensor, causing immediate water dispersion within a split second of discovery. The system then instantly notifies the user, and this notification cycle repeats every 30 seconds. This efficient procedure makes sure that user alerting, water dispersion to put out the fire, and fire detection all happen in a couple of seconds. Moreover, there is no delay in the system's response after a fire occurs, which makes our project unique. The system takes immediate action, which is crucial for containing fires and reducing their potential impact. Additionally, our project incorporated several unique features that made it stand out from other fire alarm and detection systems. We used a different servo motor as a switch to turn on the water pump, making the system faster to respond and able to water fires more quickly. We also added LED lights and a buzzer both inside and outside of the house to alert neighbors and building occupants in the event of a fire. In conclusion, the FireEye system is an excellent choice for building owners and property managers looking to improve the safety of their occupants and assets. Our project's unique features, along with the system's effectiveness and efficiency, make it a superior fire alarm and detection system.

VII. CONCLUSION AND FUTURE WORK

In conclusion, fires pose significant risks to both property and human life, making prevention and early detection crucial. Smoke and toxic fumes produced during a fire can be just as dangerous as the flames themselves, and the damage caused can be devastating. Measures such as keeping flammable materials away from heat sources, maintaining electrical appliances, and installing smoke detectors can reduce the risks associated with fires. Additionally, investing in a reliable fire alarm system, such as the FireEye system, can provide early warning and enable swift action in case of a fire, minimizing damage and protecting property and occupants. Looking to the future, We aim to develop this system even further. This could include exploring new technologies or developing more accessible and cost-effective methods for suppressing fires in existing properties along with implementing machine learning. By pursuing these avenues, we can improve overall fire safety and reduce the risks associated with fires, helping to protect both property and lives. Continuous research and development are essential to ensure that we are always adapting to new challenges and improving our ability to prevent and detect fires.

REFERENCES

- [1] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," *Phil. Trans. Roy. Soc. London*, vol. A247, pp. 529–551, April 1955.
- [2] J. Clerk Maxwell, *A Treatise on Electricity and Magnetism*, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.
- [3] Shawon, Md Sajjad Hossain, et al. "Voice controlled smart home automation system using bluetooth technology." 2021 4th International Conference on Recent Trends in Computer Science and Technology (ICRTCST). IEEE, 2022.
- [4] Tri Listyorini and Robbi Rahim, A prototype fire detection implemented using the Internet of Things and fuzzy logic, *World Trans. Eng. Technol. Educ.*, volume 16, 2018, 42-46.
- [5] Arpit Jadon, Mohd. Omama, Akshay Varshney, Mohammad Samar Ansari and Rishabh Sharma, FireNet: a specialized lightweight fire & smoke detection model for real-time IoT applications, *arXiv preprint arXiv:1905.11922*, 2019.
- [6] ABDULLAH H. ALTOWAJRI, MOHAMMED S. ALFAIFI, TARIQ A. ALSHAWI, AHMED B. IBRAHIM, and SALEH A. ALSHEBEILI, A privacy-preserving iot-based fire detector, *IEEE Access*, volume 9, 2021, 51393–51402.
- [7] Ahammed, Md Tabil, et al. "Applied Informatics in Renewable Energy Grid with Dynamic Reactive Power Configuration Based on a Probability Assessment of Transient." 2022 International Conference on Advances in Computing, Communication and Applied Informatics (ACCAI). IEEE, 2022.
- [8] M. Trinath Basu, Ragipati Karthik, J. Mahitha and V. Lokesh Reddy, IoT based forest fire detection system, *International Journal of Engineering & Technology*, volume 7, 2018, 124-126.
- [9] Kousalya, C. Gnana, et al. "Design of iot based indoor air purifier." 2022 Third International Conference on Intelligent Computing Instrumentation and Control Technologies (ICICIT). IEEE, 2022.
- [10] Ahammed, Md Tabil, Sudipto Ghosh, and Md Ashikur Rahman Ashik. "Human and Object Detection using Machine Learning Algorithm." 2022 Trends in Electrical, Electronics, Computer Engineering Conference (TEECCON). IEEE, 2022.
- [11] Giacomo Peruzzi, Alessandro Pozzebon and Mattia Van Der Meer, Fight Fire with Fire: Detecting Forest Fires with Embedded Machine Learning Models Dealing with Audio and Images on Low Power IoT Devices, *Sensors*, volume 23, 2023, 783.
- [12] Xiaogeng Ren, Chunwang Li, Xiaojun Ma, Fuxiang Chen, Haoyu Wang, Ashutosh Sharma, Gurjot Singh Gaba and Mehedi Masud, Design of multi-information fusion based intelligent electrical fire detection system for green buildings, *MDPI*, volume 13, 2021, 3405.
- [13] N N Mahzan, N I M Enzai, N M Zin and K S S K M Noh, "Design of an Arduino-based home fire alarm system with GSM module", 1st International Conference on Green and Sustainable Computing (ICoGeS), 2017.
- [14] Jenil Dosani, Nikunj Makwana, Archana Chaugule, "NodeMCU based Fire Detector System", HBRP Publication, Recent Trends in Cloud Computing and Web Engineering Volume 2 Issue 3, 2020.
- [15] [Actuator], (<https://history-computer.com/9-different-types-of-actuators-in-the-internet-of-things-iot/>), (Accessed on, June, 2023).
- [16] Ahammed, Md Tabil, et al. "Design and implementation of programmable logic controller based automatic transfer switch." *Journal of Artificial Intelligence, Machine Learning and Neural Network (JAIMLNN)* ISSN: 2799-1172 2.02 (2022): 8-18.
- [17] Khalaf, O. I., Abdulsahib, G. M., & Zghair, N. A. K. (2019). IOT fire detection system using sensor with Arduino. *AUS*, 26, 74-78.
- [18] [Sensor], (<https://www.techtarget.com/whatis/definition/sensor>), (Accessed on, June, 2023).
- [19] Ahammed, Md Tabil, et al. "Analysis of Energy Consumption for a Hybrid Green Data Center." 2022 International Conference on Innovations in Science, Engineering and Technology (ICISSET). IEEE, 2022.
- [20] Ahammed, Md Tabil, et al. "Real-time based Violence Detection from CCTV Camera using Machine Learning Method." 2022 International Conference on Industry 4.0 Technology (I4Tech). IEEE, 2022.
- [21] Rahman, Md Ashikur, et al. "Auto Signal Generate in Curvy Roads Using Object Detection." 2023 IEEE 2nd International Conference on Industrial Electronics: Developments Applications (ICIDeA). IEEE, 2023.